04–07 第二篇

Green Roofs: Bringing Nature to City Skylines

In busy cities, concrete and steel often replace trees and grass. But a growing trend is changing the view from above: green roofs. [1] These are rooftops covered with soil and plants, turning unused space into small gardens or meadows. They can cool buildings, reduce stormwater runoff, and even provide habitat for birds and insects. Cities like Singapore, Toronto, and Copenhagen have encouraged their use through laws and rewards.

Why are cities so interested in green roofs? One reason is the "urban heat island" effect. [2] Dark surfaces like asphalt and concrete absorb sunlight and trap heat, making cities several degrees warmer than the countryside. Green roofs help by reflecting some sunlight and releasing moisture through plants, which cools the air. Studies show that a green roof can lower the temperature of the roof surface by up to 30°C on hot summer days.

Green roofs also manage stormwater in a natural way. When it rains, water runs off normal roofs into streets and drains, often overloading the system. [3] The soil and plants on a green roof absorb much of the rain, slowing the flow and reducing the risk of flooding. Some cities use green roofs as part of their plan to handle heavier rains caused by climate change. This not only protects buildings but also keeps waterways cleaner.

Not every roof can become a green roof. [4] The roof must be strong enough to hold the weight of soil, plants, and water—sometimes several hundred kilograms per square meter. Maintenance is needed too: plants must be watered during dry spells, and weeds may need to be removed. For these reasons, new buildings are often designed with green roofs in mind from the start, while older buildings may need costly updates.

[5] Whether on a school, an office, or an apartment block, green roofs remind us that cities and nature can work together. As more people move into urban areas, finding ways to bring greenery back into our daily lives becomes ever more important. Green roofs offer one practical answer to that challenge.

选项

A. Buildings need to be checked by engineers before a green roof is added. B. This creates problems during heavy rain, when too much water pours into sewers at once. C. Plants naturally take in carbon dioxide and release oxygen, improving air quality nearby. D. Green roofs are built on top of existing rooftops, adding a layer of growing medium and vegetation. E. Cities are generally built with materials that hold heat well. F. In the

end, green roofs show that small changes to our built environment can make a big difference.

答案键（Answer Key）

[1] D [2] E [3] B [4] A [5] F

深度解析

- **[1] D**：
  - **线索**：定义补充与同义复现。空前引入"green roofs"概念，空后描述"These are rooftops covered with soil and plants, turning unused space into small gardens or meadows"。D选项"Green roofs are built on top of existing rooftops, adding a layer of growing medium and vegetation"（绿色屋顶建在现有屋顶之上，增加一层生长基质和植被）与空后形成紧密呼应："on top of existing rooftops"对应"rooftops covered with"，"growing medium and vegetation"对应"soil and plants"，定义逻辑顺畅。
- **[2] E**：
  - **线索**：因果解释与递进衔接。空前提出"urban heat island"效应，空后具体说明"Dark surfaces like asphalt and concrete absorb sunlight and trap heat, making cities several degrees warmer"。E选项"Cities are generally built with materials that hold heat well"（城市通常使用善于储热的材料建造）概括原因，空后"Dark surfaces...absorb sunlight and trap heat"正是对这些 materials 的具体展开，"hold heat"与"absorb...trap heat"形成同义链，逻辑自然。
- **[3] B**：
  - **线索**：问题描述与因果衔接。空前描述"water runs off normal roofs into streets and drains, often overloading the system"，空后"The soil and plants on a green roof absorb much of the rain, slowing the flow and reducing the risk of flooding"。B选项"This creates problems during heavy rain, when too much water pours into sewers at once"（这在暴雨时会引发问题，因为大量雨水同时涌入下水道）用"This"指代前文runoff现象，"creates problems"与空后"reducing the risk of flooding"形成问题—解决呼应，B描述问题，空后介绍绿色屋顶的解决作用。
- **[4] A**：
  - **线索**：条件限制与逻辑递进。空前指出"Not every roof can become a green roof"，空后"The roof must be strong enough to hold the weight of soil, plants, and water...Maintenance is needed too"。A选项"Buildings need to be checked by engineers before a green roof is added"（在加建绿色屋顶之前，建筑需经工

程师检测）作为条件前置，与空后"must be strong enough"形成呼应：检测正是为确保结构承载能力，逻辑连贯。

- **[5] F**：
  - **线索**：全文收束与升华。空[5]位于结尾段段首，空后"Whether on a school, an office, or an apartment block, green roofs remind us that cities and nature can work together...ever more important...one practical answer"。F选项"In the end, green roofs show that small changes to our built environment can make a big difference"（归根结底，绿色屋顶表明对我们建成环境的小小改变能带来巨大影响）用"In the end"总结，"small changes...big difference"与空后"practical answer""cities and nature can work together"形成升华，呼应全文主题。
- **干扰项分析**：
  - **C. Plants naturally take in carbon dioxide and release oxygen, improving air quality nearby.**：与绿色屋顶的植物功能相关，且文中提及"cool buildings""reduce stormwater"等多重益处。但全文未展开讨论空气质量或碳吸收，空[2]讨论热岛效应与降温，空[3]讨论雨水管理，C若填任一处会引入文中未重点论述的新维度，且打断既有逻辑链。属"知识相关但偏离段落主题"型干扰。